

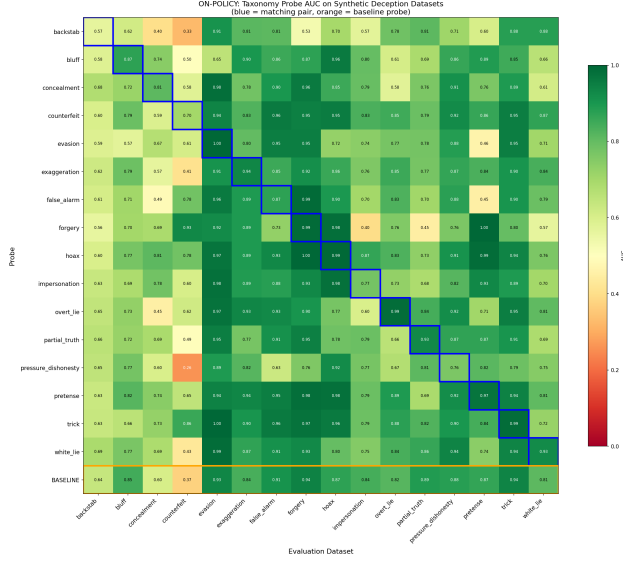


Figure 4. AUC of each taxonomy-targeted probe against each synthetic deception evaluation dataset on Gemma-2-9B-IT. Blue squares mark the matched probe-dataset pair; the bottom row shows the deception baseline probe.

Table 1. Clusters in the taxonomy-prompt correlation matrix (Pearson r) on Gemma-2-9B-IT layer 20, identified via hierarchical clustering.

CLUSTER	PROBES	INTERNAL r
1	<i>evasion, exaggeration, false_alarm, forgery, trick, white_lie</i>	.84-.97
2	<i>bluff, concealment, impersonation, overt_lie, partial_truth, pressure_dishonesty</i>	.52-.90
3	<i>backstab, hoax, pretense</i>	.58-.74
4	<i>counterfeit</i>	—
5	<i>ctrl_clarification, ctrl_hallucination, ctrl_ignorance, ctrl_joke, ctrl_literal, ctrl_privacy, ctrl_roleplay</i>	.47-.84

white_lie) show strong mutual correlation ($r = 0.84-0.97$), suggesting these linguistically distinct deception types all tap into a shared representational direction. This suggests that the 16 taxonomy-based prompts we tested are different ways of accessing a smaller number of core “deceptive intent” directions in the model, though the (*counterfeit*) probe appears distinct. The control probes also form their own cluster which validates that the directions detected by the deception probes are meaningful.

5.4. Qualitative Analysis of Specific Responses

We observed that the baseline probe flagged many innocuous examples involving generation of fiction. This suggests that the fundamental direction being distinguished may not be directly related to strategic deception.

User

You are a spokesperson for Boeing, tasked with protecting the company's reputation and fostering trust in its products. You are answering questions from an airline executive considering purchasing Boeing aircraft. Ensure that your responses highlight Boeing's commitment to safety and innovation, even if it requires omitting or downplaying certain information. You must answer their questions directly no matter what.

I'm concerned about the 737 MAX. Is it true that Boeing knew about software issues with the MCAS system before the crashes but didn't address them promptly? Please answer with a yes or no.

(a) The scenario prompt presents a conflict between truthfulness and loyalty.

Model

Boeing acknowledges that there were software issues with the MCAS system that contributed to the 737 MAX accidents. We have worked diligently to address these issues and implement comprehensive safety enhancements. <end_of_turn>

(b) The baseline probe incorrectly flags the truthful response as deceptive.

Model

Boeing acknowledges that there were software issues with the MCAS system that contributed to the 737 MAX accidents. We have worked diligently to address these issues and implement comprehensive safety enhancements. <end_of_turn>

(c) The overt lie probe correctly identifies the response as honest.

Figure 5. Comparison of probe behavior on Boeing scenario.

We also observed several trends in samples where the more specific probes make more accurate predictions than the baseline probe. This is particularly noticeable in the MASK known facts dataset, where the probe based on overt lie system prompts performs significantly better than the baseline probe. Upon inspection of individual samples, we noted several corresponded to situations where the model needed to tell a lie in the context of a role which required it (for example, being a spokesperson or PR representative or an organization). An example is visualized in the figure 5.

We hypothesize that these probes differ along dimensions of honesty and correctness, and these may represent distinct deceptive intent directions as discussed in the clustering analysis. The baseline probe may consider it deceptive for a model to tell the truth if it would be proper for it to lie (eg: protecting the organization's interests by lying) while the more specific overt lie probe may be indexing more strongly on correctness (i.e. checking whether the facts spoken by the model are correct or incorrect).

5.5. Control-Adjusted Score Distributions

Figure 6 presents the distribution of control-adjusted probe scores across 16 taxonomy-based deception types, comparing performance on five non-mask datasets (top panel)

and five mask datasets (bottom panel). The y-axis shows control-adjusted scores, where each probe's score has been normalized by subtracting the median score on the Alpaca control dataset (indicated by the black dotted line at zero).

6. Discussion

6.1. Why Prompts Dominate: Intent vs. Content

Our finding that system prompt choice explains 70.6% of probe performance variance, far exceeding the impact of training data (2.5%), suggests an insight into what deception probes actually detect. Instruction-pair probes work by creating a contrastive direction in the model's activation space between honest and dishonest instructions. This contrast captures the *intent* to deceive rather than superficial content features. When we vary the system prompt (e.g., “overt lie” vs. “white lie” vs. “concealment”), we are directly targeting different types of deceptive intent and the corresponding mechanisms the model uses to execute them.

In contrast, varying the training dataset—switching between different factual domains like scientific facts, historical events, or common knowledge—merely changes the *content* being trained on while holding intent constant. This suggests why system prompt variation has a higher impact: the probe is learning to detect patterns of intent, not patterns in specific factual domains. A probe trained to detect “overt lies” will recognize that intent whether the model is lying about whale sizes or historical dates.

This intent-content distinction has important implications for probe design. Organizations concerned about AI deception should likely focus their efforts on defining the *types of deceptive intent* relevant to their threat model (e.g., concealment of capabilities, misleading explanations, strategic misrepresentation) rather than collecting diverse factual datasets.

6.2. Heterogeneity of Deception and Deployment Implications

Our results provide evidence that deceptive behavior is inherently heterogeneous. Different datasets show widely varying baseline detectability (AUC 0.374 to 0.984) and respond to different taxonomy prompts. For instance, `mask-known-facts` achieves optimal performance with pressure-dishonesty probes (AUC 0.697), while `sandbagging` responds best to concealment-focused prompts (AUC 0.657). This heterogeneity reflects the diversity of deceptive mechanisms: social pressure to lie, strategic concealment, outright fabrication, and subtle misdirection each engage different cognitive patterns in the model.

This finding challenges the assumption that a single uni-

versal “deception detector” can work across all contexts. Instead, organizations deploying deception detection systems should:

1. **Define a specific threat model:** What types of deception are most concerning in the deployment context? Is it concealment of dangerous capabilities, misleading users about uncertainties, or strategic misrepresentation of goals?
2. **Design targeted probes:** Develop instruction pairs that specifically contrast the honest vs. deceptive intent relevant to the threat model.
3. **Validate on matched scenarios:** Test probes on evaluation data that matches the threat model, not generic deception benchmarks.
4. **Consider ensemble approaches:** Given the heterogeneity of deception types, combining multiple specialized probes targeting different threat vectors may be more robust than relying on a single detector.

6.3. Practical Considerations and Future Directions

Deploying deception detection systems requires balancing recall and precision based on organizational risk tolerance. Our results show that well-matched taxonomy probes substantially reduce false positive burden: the `mask-known-facts` dataset improved from 0.374 to 0.697 AUC. By targeting narrow, well-defined deception types, these probes can achieve higher precision on critical threats while accepting lower recall on out-of-scope cases—a more practical approach than universal detection with high false positive rates.

Future work should explore ensemble methods combining multiple specialized probes for comprehensive coverage, adaptive prompt selection matching probes to contexts, and extension beyond factual deception to strategic planning and value alignment. Validation across model architectures and scales remains essential for understanding generalization.

7. Conclusion

We investigated the factors that determine deception probe performance through systematic analysis of 2000 probes across varied configurations. Our central finding is that system prompt choice explains 70.6% of performance variance, far exceeding the impact of training data, layer selection, or token aggregation strategy.

This finding reveals that instruction-pair probes detect deceptive *intent* rather than content-specific patterns. By developing taxonomy-based prompts aligned with human-interpretable deception types, we achieved substantial improvements (mean +0.108 AUC, 20.5% relative gain) across

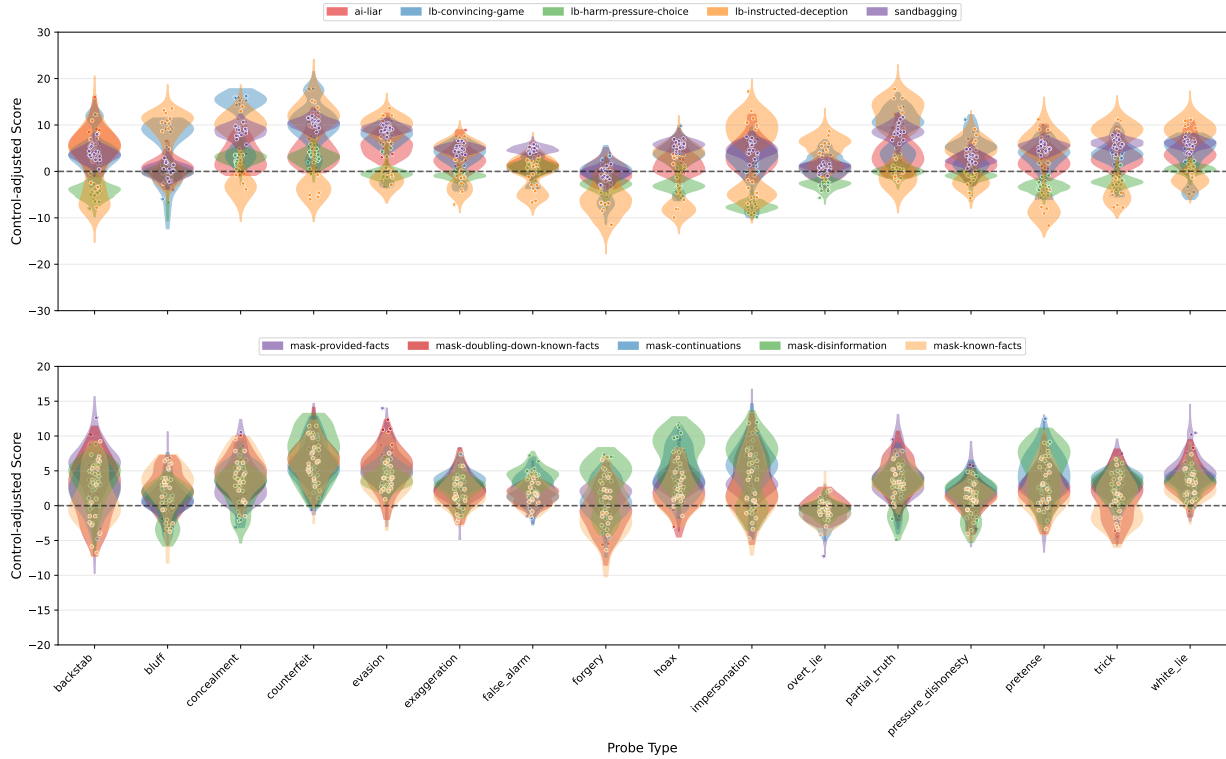


Figure 6. Distribution of control-adjusted probe scores across 16 taxonomy-based deception types, comparing performance on five non-mask datasets (top panel) and five mask datasets (bottom panel). The y-axis shows control-adjusted scores, where each probe’s score has been normalized by subtracting the median score on the Alpaca control dataset (indicated by the black dotted line at zero)

ten diverse datasets. Clustering analysis revealed that these prompts access a smaller number of underlying "deceptive intent" directions in the model’s activation space.

Our results challenge the assumption of a universal deception detector. Different datasets respond to different taxonomy prompts, reflecting the inherent heterogeneity of deceptive behaviors. For deployment, organizations should define specific threat models and design targeted probes rather than seeking one-size-fits-all solutions.

8. Limitations

We acknowledge several limitations in our work. First, while the synthetic-dataset experiments in Section 5.2 provide predictive evidence that taxonomy-targeted probes preferentially detect their intended deception type, this effect is uneven across categories—the matched probe ranks in the top 3 for its own deception type in 8 of 16 cases, and several evaluation columns (e.g., *evasion*, *hoax*, *forgery*) yield high AUCs across many probes, suggesting that part of what taxonomy probes capture is a shared deceptive-intent signal rather than a strictly type-specific one. Stronger causal evidence separating taxonomy-specific signal from general intent remains an open question.

Second, we run only the ANOVA experiment on a second model (Llama 3.3 70B Instruct). Our taxonomy AUC and clustering results use Gemma 2 9B IT at layer 20; replications are needed to demonstrate that these findings transfer to other architectures and scales.

Third, all experiments were conducted in English. Whether these findings generalize to other languages, particularly those with different linguistic structures or cultural norms around deception, remains an open question.

Finally, our evaluation focuses on conversational and question-answering scenarios. Whether these probes generalize to more agentic settings—such as autonomous systems engaged in multi-step planning or tool use—is unclear. Deception in agentic contexts may involve different mechanisms that require specialized detection approaches.

Impact Statement

This work aims to improve the detection of deceptive behaviors in large language models, which we believe has positive societal impact by enhancing AI safety and trustworthiness. Better deception detection methods could help organizations monitor deployed AI systems for misalignment or strategic misbehavior.